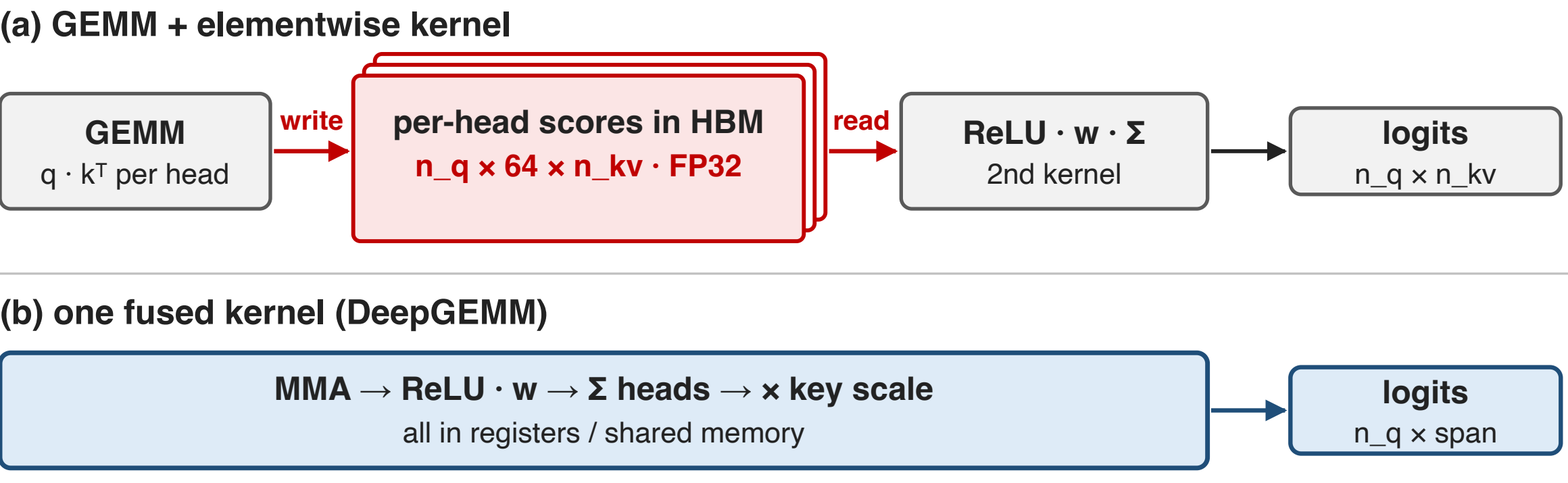
(a) GEMM + elementwise kernel
GEMM
q · kᵀ per head
write
per-head scores in HBM
n_q × 64 × n_kv · FP32
read
ReLU · w · Σ
2nd kernel
logits
n_q × n_kv
(b) one fused kernel (DeepGEMM)
MMA → ReLU · w → Σ heads → × key scale
all in registers / shared memory
logits
n_q × span